

Software Requirements Specification (SRS)

Project: Plagiarism Detection Tool using TF-IDF + Word Embeddings

1. Introduction

This project aims to develop a web-based Plagiarism Detection Tool that calculates similarity between documents using TF-IDF cosine similarity and Word Embedding similarity techniques.

2. Scope

The system allows users to upload documents (.txt, .pdf, .docx), preprocess text, compute similarity scores, highlight matched content, and generate downloadable reports.

3. Functional Requirements

- User Registration and Login
- Document Upload and Storage
- Text Preprocessing (Tokenization, Stopword Removal, Lemmatization)
- TF-IDF Similarity Computation
- Word Embedding Similarity Computation
- Similarity Report Generation
- Admin Dashboard

4. Non-Functional Requirements

- Response time under 5 seconds for documents under 5MB
- Secure authentication with password hashing
- Cloud deployment support (Render/Vercel)
- 99% uptime reliability

5. System Architecture

Client Browser → Flask Backend API → ML Similarity Engine → Database (MongoDB/PostgreSQL)

6. Database Design

Collections/Tables:

- Users (user_id, name, email, password)
- Documents (doc_id, user_id, file_name, content, upload_date)
- Results (result_id, doc1_id, doc2_id, tfidf_score, embedding_score, final_score)

7. Algorithms

TF-IDF: Convert documents to vectors and compute cosine similarity.

Word Embeddings: Generate document embeddings and compute cosine similarity.

8. Deployment

Backend deployed on Render.

Frontend deployed on Vercel.

Database hosted on MongoDB Atlas.